

SAFETY DATA SHEET

Revision Date 24-December-2021

Revision Number 5

1. Identification

Product Name Cupric Sulfate-Iodide Solution

Cat No. : NC0204004; NC1833528; XXCUSIOPP20LI

Synonyms No information available

Recommended Use Laboratory chemicals.

Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company

Importer/Distributor
Fisher Scientific
112 Colonnade Road,
Ottawa, ON K2E 7L6,
Canada
Tel: 1-800-234-7437

Manufacturer

Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number CHEMTREC®, Inside the USA: 800-424-9300
CHEMTREC®, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification

WHMIS 2015 Classification Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

Specific target organ toxicity - (repeated exposure) Category 2

Label Elements

Signal Word

Warning

Hazard Statements

May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray

Response

Get medical advice/attention if you feel unwell

Disposal

Dispose of contents/container to an approved waste disposal plant

Other Hazards

Toxic to aquatic life with long lasting effects

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Water	7732-18-5	65 - 66.5
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	9.0 - 9.5
Tripotassium citrate monohydrate	6100-05-6	8.0 - 8.5
Carbonic acid, dipotassium salt	584-08-7	7.0 - 7.5
Potassium iodide (KI)	7681-11-0	5.0
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	2.5
Sodium hydroxide	1310-73-2	< 1
Iodic acid (HIO ₃), potassium salt	7758-05-6	< 1
Potassium oxalate	583-52-8	-

4. First-aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Most important symptoms/effects	None reasonably foreseeable.
Notes to Physician	Treat symptomatically

5. Fire-fighting measures

Unsuitable Extinguishing Media	No information available
Flash Point	Not applicable
Method -	No information available

Autoignition Temperature	No information available
Explosion Limits	
Upper	No data available
Lower	No data available
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products

None known.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA

Health	Flammability	Instability	Physical hazards
2	0	0	N/A

6. Accidental release measures

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment as required.
Environmental Precautions	Do not flush into surface water or sanitary sewer system.

Methods for Containment and Clean Up Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

Storage. Keep containers tightly closed in a dry, cool and well-ventilated place.

8. Exposure controls / personal protection

Exposure Guidelines

Component	Alberta	British Columbia	Ontario TWAEV	Quebec	ACGIH TLV	OSHA PEL	NIOSH IDLH
Potassium iodide (KI)			TWA: 0.01 ppm		TWA: 0.01 ppm		
Sulfuric acid, copper(2+) salt (1:1), pentahydrate					TWA: 1 mg/m ³		IDLH: 100 mg/m ³ TWA: 1 mg/m ³
Sodium hydroxide	Ceiling: 2 mg/m ³	Ceiling: 2 mg/m ³	CEV: 2 mg/m ³	Ceiling: 2 mg/m ³	Ceiling: 2 mg/m ³	(Vacated) Ceiling: 2 mg/m ³ TWA: 2 mg/m ³	IDLH: 10 mg/m ³ Ceiling: 2 mg/m ³

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection	Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.
Hand Protection	Wear appropriate protective gloves and clothing to prevent skin exposure.

Glove material	Breakthrough time	Glove thickness	Glove comments
Natural rubber	See manufacturers	-	Splash protection only
Nitrile rubber	recommendations		
Neoprene			
PVC			

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Recommended Filter type: Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

9. Physical and chemical properties

Physical State	Liquid
Appearance	No information available
Odor	No information available
Odor Threshold	No information available
pH	
Melting Point/Range	No data available
Boiling Point/Range	No information available
Flash Point	Not applicable
Evaporation Rate	No information available
Flammability (solid,gas)	Not applicable
Flammability or explosive limits	
Upper	No data available
Lower	No data available
Vapor Pressure	No information available
Vapor Density	No information available
Specific Gravity	No information available
Solubility	No information available
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	No information available
Decomposition Temperature	No information available
Viscosity	No information available

10. Stability and reactivity

Reactive Hazard	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Incompatible products.
Incompatible Materials	Strong oxidizing agents
Hazardous Decomposition Products	None under normal use conditions
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information

Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Tripotassium citrate monohydrate	5400 mg/kg (Rat)	> 2000 mg/kg	Not listed
Carbonic acid, dipotassium salt	> 2000 mg/kg (Rat)	LD50 > 2000 mg/kg (Rabbit)	LC50 > 4.96 mg/L (Rat) 4.5 h
Potassium iodide (KI)	2779 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	Not listed
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	LD50 = 960 mg/kg (Rat)	LD50 > 8 g/kg (Rabbit)	Not listed
Sodium hydroxide	LD50 = 325 mg/kg (Rat)	LD50 = 1350 mg/kg (Rabbit)	Not listed
Iodic acid (HIO ₃), potassium salt	Not listed	LD50 > 2000 mg/kg (Rat)	Not listed
Potassium oxalate	LD50 = 660 mg/kg (Rat)	Not listed	Not listed

Toxicologically Synergistic No information available

Products

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation No information available

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS-No	IARC	NTP	ACGIH	OSHA	Mexico
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	Not listed	Not listed	Not listed	Not listed	Not listed
Tripotassium citrate monohydrate	6100-05-6	Not listed	Not listed	Not listed	Not listed	Not listed
Carbonic acid, dipotassium salt	584-08-7	Not listed	Not listed	Not listed	Not listed	Not listed
Potassium iodide (KI)	7681-11-0	Not listed	Not listed	Not listed	Not listed	Not listed
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	Not listed	Not listed	Not listed	Not listed	Not listed
Sodium hydroxide	1310-73-2	Not listed	Not listed	Not listed	Not listed	Not listed
Iodic acid (HIO ₃),	7758-05-6	Not listed	Not listed	Not listed	Not listed	Not listed

potassium salt						
Potassium oxalate	583-52-8	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known

STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and delayed No information available

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Tripotassium citrate monohydrate	NOEC = 425 mg/l, 8 days	LC50 > 10 mg/l, 24h Oncorhynchus tshawytscha	Not listed	EC50 > 50 mg/l, 48h
Carbonic acid, dipotassium salt	Not listed	LC50 < 510 mg/L/96h (Pimephales promelas)	Not listed	LC50: = 630 mg/L, 48h (Ceriodaphnia dubia)
Potassium iodide (KI)	-	Onchorhynchus mykiss: LC50: 3200 mg/L/120h	-	-
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	Not listed	Onchorhynchus mykiss: LC50 = 0.1-2.5 mg/L/96h	Photobacterium phosphoreum: EC50 = 0.25 mg/L/30min as Cu++ Photobacterium phosphoreum EC50= 1.3 mg/L/5 min as Cu++	EC50 = 0.24 mg/L/48h
Sodium hydroxide	-	LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss)	-	Not listed

Persistence and Degradability No information available

Bioaccumulation/ Accumulation No information available.

Mobility No information available.

Component	log Pow
Tripotassium citrate monohydrate	-0.2 - -1.8
Potassium iodide (KI)	0.04
Potassium oxalate	-0.81

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT	Not regulated
TDG	Not regulated
IATA	Not regulated
IMDG/IMO	Not regulated

15. Regulatory information

International Inventories

Component	CAS-No	DSL	NDSL	TSCA	TSCA Inventory notification - Active-Inactive	EINECS	ELINCS	NLP
Water	7732-18-5	X	-	X	ACTIVE	231-791-2	-	-
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	-	-	-	-	-	-	-
Tripotassium citrate monohydrate	6100-05-6	-	-	-	-	-	-	-
Carbonic acid, dipotassium salt	584-08-7	X	-	X	ACTIVE	209-529-3	-	-
Potassium iodide (KI)	7681-11-0	X	-	X	ACTIVE	231-659-4	-	-
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	-	-	-	-	-	-	-
Sodium hydroxide	1310-73-2	X	-	X	ACTIVE	215-185-5	-	-
Iodic acid (HIO ₃), potassium salt	7758-05-6	X	-	X	ACTIVE	231-831-9	-	-
Potassium oxalate	583-52-8	X	-	X	ACTIVE	209-506-8	-	-

Component	CAS-No	IECSC	KECL	ENCS	ISHL	TCSI	AICS	NZIoC	PICCS
Water	7732-18-5	X	KE-35400	X	-	X	X	X	X
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	X	-	X	-	X	X	X	X
Tripotassium citrate monohydrate	6100-05-6	X	-	X	-	X	X	X	X
Carbonic acid, dipotassium salt	584-08-7	X	KE-29083	X	X	X	X	X	X
Potassium iodide (KI)	7681-11-0	X	KE-29149	X	X	X	X	X	X
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	X	-	X	-	X	X	X	X
Sodium hydroxide	1310-73-2	X	KE-31487	X	X	X	X	X	X
Iodic acid (HIO ₃), potassium salt	7758-05-6	X	KE-29148	X	X	X	X	X	X
Potassium oxalate	583-52-8	X	KE-29170	X	X	X	X	X	X

Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

Component	Canada - National Pollutant Release Inventory (NPRI)	Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances	Canada's Chemicals Management Plan (CEPA)
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	Part 1, Group A Substance		

Legend

NPRI - National Pollutant Release Inventory

Other International Regulations

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	-	Use restricted. See item 75. (see link for restriction details)	-
Sodium hydroxide	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential	Restriction of Hazardous Substances (RoHS)
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	Not applicable	Not applicable	Not applicable	Not applicable
Tripotassium citrate monohydrate	6100-05-6	Not applicable	Not applicable	Not applicable	Not applicable
Carbonic acid, dipotassium salt	584-08-7	Listed	Not applicable	Not applicable	Not applicable
Potassium iodide (KI)	7681-11-0	Listed	Not applicable	Not applicable	Not applicable
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	Listed	Not applicable	Not applicable	Not applicable
Sodium hydroxide	1310-73-2	Listed	Not applicable	Not applicable	Not applicable
Iodic acid (HIO ₃), potassium salt	7758-05-6	Not applicable	Not applicable	Not applicable	Not applicable
Potassium oxalate	583-52-8	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Ethanedioic acid, dipotassium salt, monohydrate	6487-48-5	Not applicable	Not applicable	Not applicable	Not applicable
Tripotassium citrate monohydrate	6100-05-6	Not applicable	Not applicable	Not applicable	Not applicable
Carbonic acid, dipotassium salt	584-08-7	Not applicable	Not applicable	Not applicable	Not applicable
Potassium iodide (KI)	7681-11-0	Not applicable	Not applicable	Not applicable	Not applicable
Sulfuric acid, copper(2+) salt (1:1), pentahydrate	7758-99-8	Not applicable	Not applicable	Not applicable	Annex I - Y22
Sodium hydroxide	1310-73-2	Not applicable	Not applicable	Not applicable	Annex I - Y35
Iodic acid (HIO ₃), potassium salt	7758-05-6	Not applicable	Not applicable	Not applicable	Not applicable
Potassium oxalate	583-52-8	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information

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Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align

with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS